

CHHENG SOTHEAN

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SUMMARY

Data Science student at the Institute of Technology of Cambodia with hands-on experience in data analysis, machine learning, dashboard development, and end-to-end data pipeline projects. Skilled in Python, SQL, Power BI, PySpark, SQL Server, and data visualization. Seeking a data analyst, data engineer, or machine learning internship where I can apply technical skills to solve real business and data problems.

EDUCATION

B.Sc. Applied Mathematics & Statistics — Specialization in Data Science **2021 – Expected 2027**

Institut de Technologie du Cambodge (ITC)

Academic training covers machine learning, statistical modeling, database systems, data engineering, data visualization, probabilistic models, graph theory, artificial intelligence.

TECHNICAL SKILLS

Programming	Python, SQL, JavaScript, HTML, CSS
Data Analysis	pandas, NumPy, Matplotlib, Excel, Power BI, Jupyter Notebook
Data Engineering	PySpark, Apache Spark, SQL Server, ETL Pipeline, API Integration, CSV/JSON Processing
Machine Learning	scikit-learn, Regression, Classification, Model Evaluation, Data Preprocessing
Tools & Platforms	Git, GitHub, VS Code, SQL Server Management Studio, Power BI Desktop

PROJECT EXPERIENCE

Cambodia Air Quality Data Pipeline 2026 **2025 – 2026**

Python · PySpark · Apache Spark · SQL Server · API Integration · Power BI

- Built an end-to-end air quality data pipeline that collects live pollution data from the Open-Meteo API for five Cambodian cities, including Phnom Penh, Siem Reap, Battambang, Kampot, and Sihanoukville.
- Processed and cleaned air quality records using PySpark and Apache Spark, including PM2.5, PM10, CO, NO2, SO2, O3, AQI category, date, hour, and month features.
- Stored cleaned and aggregated datasets in SQL Server tables for city-level summaries, pollutant averages, dangerous air quality counts, and Power BI dashboard reporting.

ABA Transaction Analysis Dashboard 2026 **2025 – 2026**

Python · pandas · Regex · CSV/JSON · Power BI · Jupyter Notebook

- Built an end-to-end transaction analysis workflow parsing ABA Bank Telegram payment notifications into a structured dataset using Python, pandas, and regex, extracting transaction amount, customer name, payment method, and transaction ID.
- Analyzed 5,509 transactions with total revenue of USD79,347.90 across 3,678 unique customers and 48 payment methods, identifying customer spending patterns, daily revenue trends, transaction frequency, and monthly summaries.
- Designed an interactive Power BI dashboard to visualize revenue performance, top customers, payment method breakdowns, hourly and day-of-week transaction patterns, and business payment behavior insights.

COVID-19 EDA & Machine Learning 2026 **2025 – 2026**

Python · NumPy · pandas · scikit-learn · Streamlit · Jupyter Notebook

- Performed exploratory data analysis on COVID-19 data to understand confirmed cases, recovery patterns, death rates, active cases, and country-level trends through visual analysis.
- Cleaned and prepared the dataset for machine learning by handling missing values, selecting relevant features, transforming data, and preparing training/testing sets.
- Built and evaluated machine learning models to support COVID-19 analysis and prediction using classification or regression-based techniques.
- Developed an interactive Streamlit dashboard that allows users to explore COVID-19 statistics, view visual trends, and test model predictions through a simple web interface.

LANGUAGES

Khmer — Native **English** — Intermediate